

Data Pre-processing (Python)

5.1 Import Libraries

There are three main groups of libraries are used in this project. Including:

Category	Library
Data handling	Pandas Numpy
Visualisation	Matplotlib Seaborn Plotly.io Plotly.express
Machine learning	Sklearn sklearn.model_selection sklearn.preprocessing sklearn.ensemble sklearn.metrics

5.2 Import Data

- Listings.csv
- Reviews.csv

5.3 Data Inspection

Performed:

- Data types
- Missing values
- Duplicate records
- Feature engineering
- Outliers

5.4 Data Cleaning

5.4.1 Duplicate Records

Both of the datasets have no duplicate values.

5.4.2 Missing Values

Missing values were analysed by calculating the percentage of missing records for each dataset.

Variable	Missing (%)
name	0.06
host_since	0.06
host_location	0.30
host_response_time	46.04
host_response_rate	46.04
host_acceptance_rate	40.43
host_is_superhost	0.06
host_total_listings_count	0.06
host_has_profile_pic	0.06
host_identity_verified	0.06
District	86.77
Bedrooms	10.52
review_scores_rating	32.77
review_scores_accuracy	32.79
review_scores_cleanliness	32.77
review_scores_checkin	32.81
review_scores_communication	32.78
review_scores_location	32.81
review_scores_value	32.81

Treatment strategy

Columns with extremely high missing rates (>70%) were removed because:

- They provided limited information.
- Missing values could introduce bias.
- They reduced model reliability.

For important numerical variables:

- The name variable was excluded from further analysis because it does not provide meaningful predictive information for the model. Since the listing name is a descriptive text field rather than a measurable property characteristic, it was not considered relevant for explaining

variations in listing prices. Therefore, this variable was removed from the dataset during the feature selection stage.

- Median imputation was applied where appropriate because the median is less sensitive to extreme values than the mean. Numerical variables including bedrooms, review_scores_rating, review_scores_accuracy, review_scores_cleanliness, review_scores_checkin, review_scores_communication, review_scores_location, review_scores_value, host_total_listings_count, host_since will be filled with the median.
- Missing categorical variables were treated according to their business meaning. Binary attributes such as superhost status, identity verification, and profile picture availability were converted to False when missing, assuming the absence of confirmation indicates the feature was not available. Missing host_response_rate and host_acceptance_rate will be understood as the host did not report and will be filled it with -1 to test the correlation. Other categorical variables such as response time were labelled as "Unknown" to preserve missing information rather than introducing false assumptions.
- Missing values in host_location were imputed using the mode of host locations within each neighbourhood. This approach preserves geographical relationships by assigning missing locations based on the most common host location among similar listings rather than applying a global assumption.

5.4.3 Data Type Conversion

Boolean variables

Categorical variables will be converted into dummy variables, including host_is_superhost, host_has_profile_pic, host_identity_verified, instant_bookable'.

Currency Standardisation

Airbnb listings came from multiple countries, and the prices were stored in local currencies. To enable fair comparisons across cities, prices were converted into USD using fixed exchange rates.

Currency	Exchange rate
USD -> USD	1
EUR -> USD	1.14
AUD → USD	0.69
THB → USD	0.03
HKD → USD	0.13
ZAR → USD	0.062

TRY → USD	0.021
MXN → USD	0.057
BRL → USD	0.19

5.5 Feature Engineering

Additional variables were created to improve analysis.

5.5.1 Amenities Count

The **amenity count** variable was created by extracting the total number of available amenities from the original amenities text field. This feature was developed to measure whether additional property facilities influence Airbnb pricing, customer satisfaction, and perceived value.

The analysis aims to answer the business question:

Do listings with more amenities achieve higher prices or provide better value for customers?

The original amenities field contains a list of facilities provided by each property. This text-based information was transformed into a numerical variable representing the total number of amenities available.

Example:

The original amenities : ["Wi-Fi","Kitchen","Pool","Parking"] will be converted to Amenity Count: 4

A higher amenity count indicates that a property provides more facilities, which may contribute to:

- Higher customer satisfaction
- Better review scores
- Increased pricing capability
- Greater competitiveness in the market

The amenity count variable was used during exploratory analysis and machine learning modelling to evaluate the relationship between property facilities and Airbnb prices.

5.5.2 Host Experience

The **host experience** variable was created to measure the length of time a host has been active on Airbnb. This feature was calculated using the following formula:

Host experience = Current Year - Host since year

Host experience was included because experienced hosts may have developed better operational practices and customer management skills over time. More experienced hosts may:

- Achieve higher review ratings due to better service quality
- Receive more customer trust through established profiles
- Manage pricing strategies more effectively
- Generate higher revenue compared with newer hosts

This variable was analysed to determine whether host experience contributes to Airbnb performance indicators such as price, ratings, and customer value.

5.5.3 Value Score

To provide more meaningful insights for travellers, a **Value-for-Money index (Value score)** was developed to evaluate Airbnb listings based on the balance between guest satisfaction and accommodation price. Instead of considering price or review rating separately, this index combines both factors to identify destinations that offer the best experience relative to cost.

The Value-for-Money index was calculated by combining the average review rating and the average listing price for each city. Since review scores and prices have different scales, both variables were normalised before calculation. The index was then calculated using the following formula:

Value score = Normalised Review Rating / Normalised Price

A higher index value indicates that a city provides better accommodation value, with higher guest satisfaction achieved at a lower relative cost.

5.6 Outliers Detection

Airbnb pricing contains extreme values, which may be due to:

- Luxury properties
- Incorrect data entry
- Large accommodation capacity

Outliers were analysed to identify extreme observations that could potentially affect model performance. The Interquartile Range (IQR) method was applied to detect statistical outliers, and boxplots were used to visualise the distribution of numerical variables. The analysis identified potential outliers in variables including price, bedrooms, minimum_nights, maximum_nights, and host_total_listings_count.

However, statistical outliers were not automatically removed, as extreme values in Airbnb data may represent genuine market behaviour rather than data errors. For example, high values in price may correspond to luxury properties, while large host_total_listings_count values may represent

professional or commercial hosts managing multiple properties. These observations provide valuable information about different segments of the Airbnb market and were therefore retained.

Each variable was examined individually to distinguish between legitimate extreme observations and implausible values. Only unrealistic values, such as zero or negative values for `minimum_nights`, invalid bedroom counts, or impossible property characteristics, were considered for removal. Valid extreme observations were preserved to maintain the diversity and accuracy of the dataset.

The target variable `price` showed a highly right-skewed distribution due to a small number of high-priced listings. Instead of removing all price outliers, transformations such as logarithmic transformation were used to reduce the influence of extreme values while preserving information from premium listings.

The `minimum_nights` variable also demonstrated strong positive skewness. Although most listings had short minimum stays (e.g., one to three nights), some listings required longer stays such as 30, 90, or 180 nights. These values were considered reasonable because they may reflect long-term rental arrangements, business accommodation, or local regulations. Therefore, these observations were retained, and only impossible values were targeted for removal.

Review score variables, including `review_scores_rating`, `review_scores_accuracy`, `review_scores_cleanliness`, `review_scores_checkin`, `review_scores_communication`, `review_scores_location`, and `review_scores_value`, showed negative skewness because Airbnb guests typically provide high ratings. These values were not considered problematic because the distribution reflects normal user behaviour, where most satisfactory listings receive ratings close to the maximum score.

Overall, the outlier treatment strategy focused on understanding the business context of each variable rather than removing observations solely based on statistical thresholds. This approach ensured that meaningful market patterns, including luxury properties, professional hosts, and long-term rentals, were preserved while reducing the impact of potential data errors.